

UNIVERSITY OF CENTRAL PUNJAB (UCP)

COURSE TYPES CLASSIFICATION DOCUMENT

Academic Year 2026 (RAG Dataset Version)

1. INTRODUCTION

The University of Central Punjab (UCP) follows a structured credit-hour system where every degree is built using different categories of courses. Each course type serves a specific academic purpose, such as building core knowledge, specialization, or general education.

This document explains all course categories used across UCP degree programs.

2. CORE COURSES (C)

2.1 Definition

Core courses are **mandatory subjects** that every student of a degree program must complete successfully in order to graduate.

2.2 Characteristics

- Compulsory for all students in a program
 - Cannot be replaced or substituted
 - Directly related to the major degree
 - Must be passed to progress in degree
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2.3 Purpose

- Build foundational knowledge
 - Develop essential technical skills
 - Ensure academic standardization
 - Prepare for advanced courses
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2.4 Examples (Computer Science)

- Programming Fundamentals
 - Object Oriented Programming
 - Data Structures & Algorithms
 - Operating Systems
 - Database Systems
 - Computer Networks
 - Software Engineering
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2.5 Examples (Business)

- Principles of Management
 - Financial Accounting
 - Marketing Fundamentals
 - Business Communication
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2.6 Examples (Engineering)

- Circuit Analysis
 - Engineering Mechanics
 - Thermodynamics
 - Signals and Systems
-

3. ELECTIVE COURSES (E)

3.1 Definition

Elective courses are **optional subjects** that students choose based on interest or specialization.

3.2 Characteristics

- Student choice-based
 - Limited from a predefined list
 - Used for specialization
 - May vary each semester
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3.3 Purpose

- Allow academic flexibility
 - Support specialization paths
 - Improve career alignment
 - Enhance skill diversity
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3.4 Examples

- Machine Learning
 - Cyber Security
 - Mobile App Development
 - Digital Marketing
 - Entrepreneurship
 - Cloud Computing
 - Artificial Intelligence
-

4. UNIVERSITY ELECTIVES (UE)

4.1 Definition

University Electives are **general education courses required for all degree programs**, regardless of major.

4.2 Characteristics

- Mandatory for all students
 - Non-technical or soft skill oriented
 - Improve communication and ethics
 - Standard across university
-

4.3 Purpose

- Develop communication skills
 - Build ethical understanding
 - Enhance social awareness
 - Improve soft skills
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4.4 Common UE Courses

- English Communication Skills
 - Technical Writing
 - Islamic Studies / Ethics
 - Pakistan Studies
 - Communication Skills
 - Sociology Basics
 - Entrepreneurship Fundamentals
-

5. CS / PROGRAM ELECTIVES (CS-E / PE)

5.1 Definition

Program Electives are **specialized courses related to a specific degree program**, especially Computer Science and IT-related fields.

5.2 Characteristics

- Domain-specific electives
 - Higher-level advanced topics
 - Required for specialization tracks
 - More technical in nature
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5.3 CS Elective Examples

- Machine Learning
 - Deep Learning
 - Artificial Intelligence
 - Natural Language Processing
 - Computer Vision
 - Cyber Security
 - Cloud Computing
 - Game Development
 - Mobile Application Development
 - Data Science
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5.4 Purpose

- Advanced skill development
 - Industry readiness
 - Research preparation
 - Specialization in emerging fields
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6. SUPPORTING COURSES (SC)

6.1 Definition

Supporting courses are **foundation subjects that support the main degree program**, especially in math, science, and analytical thinking.

6.2 Characteristics

- Mandatory in early semesters

- Build analytical foundation
 - Pre-requisite for core courses
 - Focus on mathematics and logic
-

6.3 Examples

- Calculus
 - Linear Algebra
 - Probability & Statistics
 - Discrete Mathematics
 - Physics
 - Applied Mathematics
 - Numerical Computing
-

6.4 Purpose

- Strengthen problem-solving ability
 - Support technical courses
 - Build mathematical foundation
 - Enable AI/CS understanding
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7. FREE ELECTIVES (FE)

7.1 Definition

Free Electives allow students to select courses **outside their department**.

7.2 Characteristics

- Cross-disciplinary
 - Optional choice
 - Broadens knowledge
 - Limited credit hours allowed
-

7.3 Examples

- Psychology Basics
 - Business Fundamentals
 - Entrepreneurship
 - Sociology
 - Economics
 - Design Thinking
-

8. COURSE TYPE COMPARISON TABLE

Type	Mandatory	Purpose	Example
Core (C)	Yes	Degree foundation	Data Structures
Elective (E)	No	Specialization	AI, ML
University Elective (UE)	Yes	Soft skills	English
CS Elective (CS-E)	No	Advanced CS topics	Deep Learning
Supporting (SC)	Yes	Foundational knowledge	Math
Free Elective (FE)	No	Cross-discipline learning	Psychology

9. DEGREE STRUCTURE USAGE

Each UCP degree uses a combination of:

- Core Courses (C)
 - Supporting Courses (SC)
 - University Electives (UE)
 - Program Electives (PE / CS-E)
 - Free Electives (FE)
-

10. CREDIT DISTRIBUTION MODEL

Typical undergraduate degree:

- Core Courses: 50–60%
- Supporting Courses: 15–20%
- Electives: 15–25%

- University Electives: 5–10%
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11. ROADMAP LOGIC FOR STUDENTS

First Year

- Supporting Courses + Basic Core

Second Year

- Core + Programming + Foundations

Third Year

- Advanced Core + Electives begin

Final Year

- Specialization + Final Year Project
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UNIVERSITY OF CENTRAL PUNJAB (UCP)

BACHELOR OF SCIENCE IN COMPUTER SCIENCE (BSCS)

COURSE CATALOG & DEGREE ROADMAP (DETAILED 2026)

1. PROGRAM OVERVIEW

The Bachelor of Science in Computer Science (BSCS) at UCP is a **4-year degree program** designed to provide students with strong foundations in computing, programming, algorithms, artificial intelligence, and software systems.

Program Objectives

Students will be able to:

- Develop software applications
 - Understand algorithms and data structures
 - Work with databases and operating systems
 - Build AI and machine learning systems
 - Design scalable software solutions
 - Solve real-world computing problems
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Duration

- 4 Years
 - 8 Semesters
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Total Credit Hours

- 124 – 136 Credit Hours
-

Final Year Requirement

- Final Year Project (FYP) Mandatory
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2. COURSE STRUCTURE BREAKDOWN

BSCS degree consists of:

- Core Courses (C)
 - Supporting Courses (SC)
 - University Electives (UE)
 - CS Electives (CS-E)
 - Free Electives (FE)
-

3. YEAR 1 – FOUNDATION LEVEL

Semester 1

- Programming Fundamentals (C)
 - Introduction to ICT (SC)
 - English Communication (UE)
 - Calculus I (SC)
 - Pakistan Studies (UE)
-

Semester 2

- Object Oriented Programming (C)
- Digital Logic Design (SC)
- Discrete Mathematics (SC)
- Technical Writing (UE)

- Islamic Studies / Ethics (UE)
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Skills Learned

- Basic programming (C++ / Python)
 - Problem solving
 - Mathematical thinking
 - Logic building
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4. YEAR 2 – CORE DEVELOPMENT

Semester 3

- Data Structures & Algorithms (C)
 - Database Systems (C)
 - Probability & Statistics (SC)
 - Computer Organization & Assembly (SC)
 - Communication Skills (UE)
-

Semester 4

- Operating Systems (C)
 - Software Engineering (C)
 - Web Development (C/PE)
 - Linear Algebra (SC)
 - Theory of Automata (C)
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Skills Learned

- Data structures mastery
 - Database design
 - Web development basics
 - System-level programming
-

5. YEAR 3 – ADVANCED COMPUTER SCIENCE

Semester 5

- Computer Networks (C)
 - Artificial Intelligence (CS-E)
 - Mobile Application Development (CS-E)
 - Numerical Computing (SC)
 - Research Methodology (UE)
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Semester 6

- Machine Learning (CS-E)
 - Information Security (CS-E)
 - Cloud Computing (CS-E)
 - Software Project Management (C)
 - Human Computer Interaction (HCI) (CS-E)
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Skills Learned

- AI & ML fundamentals
 - Networking concepts
 - Cloud systems
 - Security awareness
 - Mobile app development
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6. YEAR 4 – SPECIALIZATION & FINAL PROJECT

Semester 7

- Deep Learning (CS-E)
- Big Data Analytics (CS-E)

- Compiler Construction (C)
 - Elective II (CS-E)
 - Final Year Project Phase I (C)
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Semester 8

- Final Year Project Phase II (C)
 - Advanced Elective III (CS-E)
 - Ethics in Computing (UE)
 - Entrepreneurship (UE)
-

Skills Learned

- Industry-level project development
 - AI specialization
 - Research-based learning
 - System design
-

7. CORE COURSES (SUMMARY)

- Programming Fundamentals
 - OOP
 - Data Structures
 - Algorithms
 - Database Systems
 - Operating Systems
 - Computer Networks
 - Software Engineering
 - Theory of Automata
 - Compiler Construction
 - Final Year Project
-

8. CS ELECTIVES (SPECIALIZATION OPTIONS)

Students choose electives from:

Artificial Intelligence Track

- Machine Learning
- Deep Learning
- Neural Networks
- NLP
- Computer Vision

Software Development Track

- Mobile App Development
- Web Engineering
- UI/UX Design
- DevOps

Data Science Track

- Big Data Analytics
- Data Mining
- Data Engineering

Security Track

- Cyber Security
- Ethical Hacking
- Network Security

Cloud & Systems Track

- Cloud Computing
- Distributed Systems
- Parallel Computing

9. UNIVERSITY ELECTIVES (UE)

- English Communication
- Technical Writing
- Islamic Studies / Ethics
- Pakistan Studies

- Entrepreneurship
 - Communication Skills
-

10. SUPPORTING COURSES (SC)

- Calculus I & II
 - Discrete Mathematics
 - Probability & Statistics
 - Linear Algebra
 - Physics (Basic Level)
 - Numerical Methods
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11. FREE ELECTIVES (FE)

Students may choose from:

- Psychology
 - Business Basics
 - Economics
 - Sociology
 - Design Thinking
 - Management Studies
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12. FINAL YEAR PROJECT (FYP)

Requirements

Students must:

- Identify real-world problem
 - Design system architecture
 - Develop working solution
 - Write technical report
 - Present viva defense
-

Example Projects

- AI Chatbot
 - E-commerce System
 - Hospital Management System
 - Smart Attendance System
 - Crop Prediction System
-

13. SKILL ROADMAP

Year 1

- Logic building
- Basic programming

Year 2

- Data structures + databases

Year 3

- AI + systems + networking

Year 4

- Specialization + project building
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14. CAREER PATHS

Graduates can work as:

- Software Engineer
- AI Engineer
- Data Scientist
- Web Developer
- Mobile App Developer
- Cyber Security Analyst
- Cloud Engineer

- Research Assistant
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15. INDUSTRY ALIGNMENT

The BSCS program is aligned with:

- Software Houses
 - AI Industry
 - Banking Sector
 - Telecom Industry
 - Government IT Projects
 - Freelancing Market
-

UNIVERSITY OF CENTRAL PUNJAB (UCP)

BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING (BSSE)

COURSE CATALOG & DEGREE ROADMAP (DETAILED 2026)

1. PROGRAM OVERVIEW

The BS Software Engineering (BSSE) program at UCP is a **4-year professional degree** focused on designing, developing, testing, and maintaining high-quality software systems using engineering principles.

Program Goals

Students will learn to:

- Design large-scale software systems
 - Apply software development lifecycle (SDLC)
 - Build web, mobile, and enterprise applications
 - Work in Agile and DevOps environments
 - Manage software projects efficiently
 - Ensure software quality and testing
-

Duration

- 4 Years
 - 8 Semesters
-

Credit Hours

- 124 – 136 Credit Hours
-

Final Requirement

- Final Year Project (FYP) Mandatory
-

2. COURSE STRUCTURE

BSSE includes:

- Core Courses (C)
 - Software Engineering Core (SE-C)
 - Supporting Courses (SC)
 - University Electives (UE)
 - Software Electives (SE-E)
 - Free Electives (FE)
-

3. YEAR 1 – FOUNDATION

Semester 1

- Programming Fundamentals (C)
 - Introduction to ICT (SC)
 - English Communication (UE)
 - Calculus I (SC)
 - Pakistan Studies (UE)
-

Semester 2

- Object Oriented Programming (C)
- Discrete Mathematics (SC)
- Digital Logic Design (SC)

- Technical Writing (UE)
 - Islamic Studies / Ethics (UE)
-

Skills Learned

- Programming basics
 - Problem solving
 - Logical thinking
 - Mathematical foundation
-

4. YEAR 2 – SOFTWARE FOUNDATION

Semester 3

- Data Structures & Algorithms (C)
 - Database Systems (C)
 - Software Engineering Fundamentals (SE-C)
 - Probability & Statistics (SC)
 - Communication Skills (UE)
-

Semester 4

- Software Requirements Engineering (SE-C)
 - Operating Systems (C)
 - Web Engineering (SE-C)
 - Linear Algebra (SC)
 - Software Design & Architecture (SE-C)
-

Skills Learned

- Software lifecycle understanding
- System design basics
- Database integration
- Web development fundamentals

5. YEAR 3 – ADVANCED SOFTWARE ENGINEERING

Semester 5

- Software Quality Assurance (SE-C)
- Software Project Management (SE-C)
- Computer Networks (C)
- UI/UX Design (SE-E)
- Research Methodology (UE)

Semester 6

- Mobile Application Development (SE-E)
- DevOps Fundamentals (SE-E)
- Cloud Computing (SE-E)
- Information Security (SE-E)
- Human Computer Interaction (HCI) (SE-E)

Skills Learned

- Agile methodologies
- Testing & QA
- Cloud & DevOps
- Mobile development
- Security awareness

6. YEAR 4 – SPECIALIZATION & FINAL PROJECT

Semester 7

- Advanced Software Architecture (SE-C)
 - Software Process Improvement (SE-E)
 - Elective I (SE-E)
 - Final Year Project Phase I (C)
 - Entrepreneurship (UE)
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Semester 8

- Final Year Project Phase II (C)
 - Elective II (SE-E)
 - Ethics in Software Engineering (UE)
 - Industrial Training / Internship
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Skills Learned

- Industry-level software development
 - Project deployment
 - System optimization
 - Professional teamwork
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7. CORE COURSES (C)

- Programming Fundamentals
 - Object Oriented Programming
 - Data Structures
 - Algorithms
 - Database Systems
 - Operating Systems
 - Computer Networks
-

8. SOFTWARE ENGINEERING CORE (SE-C)

- Software Engineering Fundamentals

- Requirements Engineering
 - Software Design & Architecture
 - Software Testing Concepts
 - Software Project Management
 - Software Quality Assurance
-

9. SOFTWARE ELECTIVES (SE-E)

Students select specialization electives:

Development Track

- Web Engineering
- Mobile App Development
- Full Stack Development

DevOps Track

- DevOps Practices
- Cloud Infrastructure
- CI/CD Pipelines

Security Track

- Software Security
- Cyber Security Basics
- Secure Coding

UI/UX Track

- Human Computer Interaction
 - UI/UX Design
 - Product Design
-

10. UNIVERSITY ELECTIVES (UE)

- English Communication
- Technical Writing

- Islamic Studies / Ethics
 - Pakistan Studies
 - Entrepreneurship
 - Communication Skills
-

11. SUPPORTING COURSES (SC)

- Calculus I & II
 - Discrete Mathematics
 - Linear Algebra
 - Probability & Statistics
 - Digital Logic Design
-

12. FREE ELECTIVES (FE)

Optional cross-department courses:

- Business Management
 - Psychology
 - Economics
 - Sociology
 - Design Thinking
-

13. FINAL YEAR PROJECT (FYP)

Requirements

Students must:

- Identify real-world problem
 - Design software architecture
 - Develop full system
 - Test and deploy application
 - Submit technical report
 - Present viva demonstration
-

Example Projects

- Hospital Management System
 - E-commerce Platform
 - School Management System
 - AI Chatbot System
 - Inventory Management System
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14. ROADMAP SUMMARY

Year 1

- Programming + logic building

Year 2

- Core software engineering foundations

Year 3

- QA, DevOps, mobile + cloud systems

Year 4

- Advanced architecture + FYP
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15. CAREER PATHS

Graduates can become:

- Software Engineer
- Full Stack Developer
- DevOps Engineer
- QA Engineer
- System Analyst
- Mobile App Developer
- Cloud Engineer
- Software Architect

16. INDUSTRY ALIGNMENT

Program is aligned with:

- Software Houses
 - IT Industry
 - Freelancing Market
 - Banking Software Systems
 - Enterprise Applications
-

UNIVERSITY OF CENTRAL PUNJAB (UCP)

BACHELOR OF SCIENCE IN DATA SCIENCE (BSDS)

COURSE CATALOG & DEGREE ROADMAP (DETAILED 2026)

1. PROGRAM OVERVIEW

The BS Data Science (BSDS) program at UCP is a **4-year interdisciplinary degree** focused on extracting knowledge from data using statistics, programming, machine learning, and artificial intelligence.

The program combines:

- Computer Science
 - Mathematics
 - Statistics
 - Artificial Intelligence
 - Data Engineering
-

Program Objectives

Students will be able to:

- Analyze large datasets
 - Build predictive models
 - Apply machine learning algorithms
 - Visualize and interpret data
 - Work with big data systems
 - Solve real-world business problems using data
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Duration

- 4 Years
 - 8 Semesters
-

Credit Hours

- 124 – 136 Credit Hours
-

Final Requirement

- Final Year Project (FYP) Mandatory
-

2. COURSE STRUCTURE

BSDS includes:

- Core Courses (C)
 - Data Science Core (DS-C)
 - Supporting Courses (SC)
 - University Electives (UE)
 - Data Science Electives (DS-E)
 - Free Electives (FE)
-

3. YEAR 1 – FOUNDATIONS

Semester 1

- Programming Fundamentals (C)
- Calculus I (SC)
- English Communication (UE)
- Introduction to ICT (SC)
- Pakistan Studies (UE)

Semester 2

- Object Oriented Programming (C)
- Discrete Mathematics (SC)
- Linear Algebra (SC)
- Technical Writing (UE)
- Islamic Studies / Ethics (UE)

Skills Learned

- Programming basics (Python/C++)
- Mathematical foundations
- Logical thinking
- Problem solving

4. YEAR 2 – DATA FOUNDATIONS

Semester 3

- Data Structures & Algorithms (C)
- Probability & Statistics (SC)
- Database Systems (C)
- Data Analysis with Python (DS-C)
- Communication Skills (UE)

Semester 4

- Data Visualization (DS-C)
 - Machine Learning Fundamentals (DS-C)
 - Numerical Computing (SC)
 - Operating Systems (C)
 - Research Methodology (UE)
-

Skills Learned

- Data handling
 - Python for analytics
 - Basic ML models
 - Visualization tools (Matplotlib, Power BI)
-

5. YEAR 3 – MACHINE LEARNING & BIG DATA

Semester 5

- Machine Learning (DS-C)
 - Big Data Analytics (DS-E)
 - Data Mining (DS-E)
 - Computer Networks (C)
 - Data Engineering Fundamentals (DS-C)
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Semester 6

- Deep Learning (DS-E)
 - Artificial Intelligence (DS-E)
 - Data Warehousing (DS-E)
 - Cloud Computing (DS-E)
 - Statistical Modeling (DS-C)
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Skills Learned

- ML algorithms
 - Big data processing
 - Cloud-based data systems
 - Neural networks
-

6. YEAR 4 – ADVANCED DATA SCIENCE

Semester 7

- Advanced Machine Learning (DS-E)
 - Natural Language Processing (DS-E)
 - Computer Vision (DS-E)
 - Data Security & Ethics (DS-C)
 - Final Year Project Phase I (C)
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Semester 8

- Final Year Project Phase II (C)
 - Data Science Capstone Project (DS-E)
 - Entrepreneurship (UE)
 - Ethics in Computing (UE)
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Skills Learned

- AI applications
 - Real-world data systems
 - Research-based projects
 - Industry deployment
-

7. CORE COURSES (C)

- Programming Fundamentals
 - Object Oriented Programming
 - Data Structures
 - Algorithms
 - Database Systems
 - Operating Systems
 - Computer Networks
-

8. DATA SCIENCE CORE (DS-C)

- Data Analysis with Python
 - Data Visualization
 - Machine Learning Fundamentals
 - Statistical Modeling
 - Data Engineering Basics
-

9. DATA SCIENCE ELECTIVES (DS-E)

Students choose specialization tracks:

AI & Deep Learning Track

- Deep Learning
- Neural Networks
- Computer Vision
- NLP

Big Data Track

- Big Data Analytics
- Data Warehousing
- Hadoop/Spark Concepts

Business Intelligence Track

- Data Mining
- Business Analytics
- Decision Support Systems

Advanced Analytics Track

- Time Series Analysis
 - Predictive Modeling
 - Advanced Statistics
-

10. UNIVERSITY ELECTIVES (UE)

- English Communication
 - Technical Writing
 - Islamic Studies / Ethics
 - Pakistan Studies
 - Entrepreneurship
 - Communication Skills
-

11. SUPPORTING COURSES (SC)

- Calculus I & II
 - Linear Algebra
 - Probability & Statistics
 - Discrete Mathematics
 - Numerical Methods
-

12. FREE ELECTIVES (FE)

Optional cross-disciplinary subjects:

- Psychology
 - Business Management
 - Economics
 - Sociology
 - Marketing Basics
 - Design Thinking
-

13. FINAL YEAR PROJECT (FYP)

Requirements

Students must:

- Identify data-driven problem
- Collect and preprocess data

- Apply ML/AI techniques
 - Build predictive system
 - Evaluate model performance
 - Present final report
-

Example Projects

- Stock Market Prediction System
 - Crop Yield Prediction System
 - Disease Detection using ML
 - Customer Churn Prediction
 - AI Chatbot using NLP
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14. ROADMAP SUMMARY

Year 1

- Programming + math foundations

Year 2

- Data handling + visualization + basic ML

Year 3

- Machine learning + AI + big data

Year 4

- Advanced AI + real-world projects
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15. CAREER PATHS

Graduates can become:

- Data Scientist

- Machine Learning Engineer
 - Data Analyst
 - AI Engineer
 - Business Intelligence Analyst
 - Data Engineer
 - Research Analyst
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16. INDUSTRY ALIGNMENT

This program is aligned with:

- AI Industry
- Banking & Finance Sector
- E-commerce platforms
- Healthcare analytics
- Government data systems
- Freelancing (Upwork/Fiverr)

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□ UNIVERSITY OF CENTRAL PUNJAB (UCP)

BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE (BSAI)

COURSE CATALOG & DEGREE ROADMAP (DETAILED 2026)

1. PROGRAM OVERVIEW

The BS Artificial Intelligence (BSAI) program is a **4-year advanced computing degree** focused on designing intelligent systems capable of learning, reasoning, perception, and decision-making.

It combines:

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Robotics
- Data Science
- Computer Vision

Program Objectives

Students will be able to:

- Build AI-based applications
- Train machine learning models
- Develop intelligent systems
- Work with neural networks
- Design autonomous systems

- Solve real-world AI problems
-

Duration

- 4 Years
 - 8 Semesters
-

Credit Hours

- 124 – 136 Credit Hours
-

Final Requirement

- Final Year Project (AI-based mandatory)
-

2. COURSE STRUCTURE

Includes:

- Core Courses (C)
 - AI Core (AI-C)
 - Supporting Courses (SC)
 - University Electives (UE)
 - AI Electives (AI-E)
 - Free Electives (FE)
-

3. YEAR 1 – FOUNDATIONS

Semester 1

- Programming Fundamentals (C)
- Calculus I (SC)

- Introduction to ICT (SC)
 - English Communication (UE)
 - Pakistan Studies (UE)
-

Semester 2

- Object Oriented Programming (C)
 - Discrete Mathematics (SC)
 - Linear Algebra (SC)
 - Technical Writing (UE)
 - Islamic Studies / Ethics (UE)
-

Skills Learned

- Programming basics
 - Mathematical reasoning
 - Logical thinking
 - Problem solving
-

4. YEAR 2 – AI FOUNDATIONS

Semester 3

- Data Structures & Algorithms (C)
 - Probability & Statistics (SC)
 - Database Systems (C)
 - Introduction to Artificial Intelligence (AI-C)
 - Communication Skills (UE)
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Semester 4

- Machine Learning Fundamentals (AI-C)
- Numerical Computing (SC)
- Operating Systems (C)
- Data Visualization (AI-C)

- Research Methodology (UE)
-

Skills Learned

- Basic AI concepts
 - Data handling
 - ML basics
 - Python AI libraries
-

5. YEAR 3 – ADVANCED AI SYSTEMS

Semester 5

- Machine Learning (AI-C)
 - Neural Networks (AI-C)
 - Computer Vision (AI-E)
 - Computer Networks (C)
 - AI Ethics (AI-C)
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Semester 6

- Deep Learning (AI-E)
 - Natural Language Processing (AI-E)
 - Robotics (AI-E)
 - Cloud Computing for AI (AI-E)
 - Software Engineering (C)
-

Skills Learned

- Deep learning models
- NLP systems
- Vision-based AI
- Robotics fundamentals
- AI deployment

6. YEAR 4 – SPECIALIZATION & RESEARCH

Semester 7

- Advanced AI Systems (AI-E)
- Reinforcement Learning (AI-E)
- Big Data Analytics (AI-E)
- Final Year Project Phase I (C)
- Entrepreneurship (UE)

Semester 8

- Final Year Project Phase II (C)
- AI Capstone Project
- Ethics in Computing (UE)
- Industrial Internship

Skills Learned

- Industry-level AI systems
- Research implementation
- Production AI deployment
- Advanced model training

7. CORE COURSES (C)

- Programming Fundamentals
- Object Oriented Programming
- Data Structures
- Algorithms
- Database Systems
- Operating Systems
- Computer Networks

8. AI CORE COURSES (AI-C)

- Introduction to Artificial Intelligence
- Machine Learning Fundamentals
- Neural Networks Basics
- Data Visualization for AI
- AI Ethics

9. AI ELECTIVES (AI-E)

Students specialize in:

Deep Learning Track

- Deep Learning
- Reinforcement Learning
- Generative AI

Vision Track

- Computer Vision
- Image Processing
- Pattern Recognition

NLP Track

- Natural Language Processing
- Text Mining
- Speech Recognition

Robotics Track

- Robotics Systems
 - Autonomous Agents
 - Control Systems
-

10. UNIVERSITY ELECTIVES (UE)

- English Communication
 - Technical Writing
 - Islamic Studies / Ethics
 - Pakistan Studies
 - Entrepreneurship
-

11. SUPPORTING COURSES (SC)

- Calculus I & II
 - Linear Algebra
 - Probability & Statistics
 - Discrete Mathematics
 - Numerical Methods
-

12. FINAL YEAR PROJECT (FYP)

Requirements

Students must:

- Build AI-based system
 - Train ML/DL models
 - Evaluate performance
 - Deploy working prototype
 - Present research report
-

Example Projects

- AI Chatbot
- Face Recognition System
- Disease Detection AI
- Smart Surveillance System
- Recommendation Engine

13. CAREER PATHS

Graduates can work as:

- AI Engineer
- Machine Learning Engineer
- Data Scientist
- Computer Vision Engineer
- NLP Engineer
- Robotics Engineer

14. INDUSTRY ALIGNMENT

Aligned with:

- AI Industry
- Robotics Industry
- Healthcare AI
- FinTech AI
- Automation Systems
- Research Labs

📄 ELECTIVES MAP DOCUMENT (UCP)

COURSE SELECTION & ELECTIVE STRUCTURE GUIDE

1. INTRODUCTION

This document defines how students select electives across all UCP degree programs, including CS, SE, DS, and AI.

It helps students and AI systems understand:

- Which electives belong to which category
 - How electives are grouped
 - Restrictions and dependencies
 - Specialization paths
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2. ELECTIVE TYPES SUMMARY

Type	Full Form	Description
CS-E	Computer Science Electives	Advanced CS subjects
SE-E	Software Engineering Electives	Software design & DevOps
AI-E	Artificial Intelligence Electives	AI/ML specialization
DS-E	Data Science Electives	Data & analytics subjects
UE	University Electives	Soft skills & general courses
FE	Free Electives	Cross-department courses
SC	Supporting Courses	Math & foundation subjects

3. COMPUTER SCIENCE ELECTIVES (CS-E)

- Machine Learning
 - Deep Learning
 - Cyber Security
 - Cloud Computing
 - Mobile App Development
 - Game Development
 - NLP
 - Computer Vision
 - Blockchain Basics
 - Distributed Systems
-

4. SOFTWARE ENGINEERING ELECTIVES (SE-E)

- DevOps
 - Agile Development
 - Software Testing Automation
 - UI/UX Design
 - Cloud Engineering
 - Software Architecture
 - CI/CD Pipelines
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5. ARTIFICIAL INTELLIGENCE ELECTIVES (AI-E)

- Reinforcement Learning
 - Neural Networks
 - Computer Vision
 - NLP
 - Generative AI
 - Robotics
 - Intelligent Agents
-

6. DATA SCIENCE ELECTIVES (DS-E)

- Big Data Analytics
 - Data Mining
 - Data Engineering
 - Predictive Modeling
 - Time Series Analysis
 - Business Intelligence
 - Advanced Statistics
-

7. UNIVERSITY ELECTIVES (UE)

Common for all programs:

- English Communication
 - Technical Writing
 - Islamic Studies / Ethics
 - Pakistan Studies
 - Entrepreneurship
 - Communication Skills
-

8. FREE ELECTIVES (FE)

Students may choose from other departments:

- Psychology
 - Economics
 - Sociology
 - Marketing
 - Management
 - Design Thinking
 - Digital Media Basics
-

9. ELECTIVE RULES

- CS students must take minimum 3–6 CS electives
- AI students must take AI-focused electives

- SE students must prioritize SE electives
 - DS students must take ML + statistics electives
 - UE courses are mandatory for all programs
-

10. ELECTIVE SELECTION LOGIC (FOR CHATBOT)

If user asks:

“Which electives should I take in CS?”

Return:

- ML
 - AI
 - Cyber Security
 - Cloud Computing
 - NLP
 - Mobile App Dev
-

If user asks:

“What is difference between CS-E and UE?”

Explain:

- CS-E = technical specialization
- UE = general education